MIL-C-21850C(NU)
13 December 1979
SUPERSEDING
MIL-C-21850B(SA)
12 March 1975

MILITARY SPECIFICATION

COVERALLS, CATAPULT, CREWMEN'S

This specification is approved for use by Navy Clothing and Textile Research Facility, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a utility coverall fabricated from wind resistant cotton and nylon blend sateen fabric.

1.2 Classification. The coverals shall be of one type only and in the following sizes (see 6.2):

Schedule of Sizes

Small (36-38)	Medium (40-42)	Large (44-46)

2. APPLICABLE DOCUMENTS

2.1 Issue of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-290	- Nickel Plating (Electrodeposited)
DDD-L-20	- Label; For Clothing, Equipage and Tentage (General Use)
DDD-T-86	- Tape, Textile, Cotton, General Purpose (Unbleached, Bleached or Dyed)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposel (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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MIL-B-371	- Braid, Textile Tubular
MIL-E-480	- Enamel, Baking, Phenol or Urea-Formaldehyde
MIL-F-495	- Finish, Chemical, Black, for Copper Alloys
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-W-5664	- Webbing, Textile Elastic, Cotton
MIL-B-17757	- Boxes Shipping, Fiberboard, (Modular Sizes)
MIL-E-20652	- Eyelets, Metallic, and Eyelet Washers, Metallic; General Specification for
MIL-F-21840	- Fastener Tape, Hook and Pile, Synthetics
MIL-C-43191	- Cloth, Wind Resistant Sateen, Cotton and Nylon
MIL-T-43548	- Thread, Polyester, Cotton-Covered

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-601	- Rubber, Sampling and Testing
FED-STD-751	- Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection for Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1668	- Provisions for Evaluating Quality of Cloth Coveralls
MS-27980	- Fasteners, Snap, Style 2 (Regular Wire Spring Clamp Type)

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U. S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCICATION, INC., AGENT
National Motor Freight Classification

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(Application for copies should be addressed to American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W. Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, ILL 60606.)

AMERICAN SOCIETY FOR TESTING MATERIALS STANDARDS
D-1056 - Tests for Sponge and Expanded Cellular Rubber Products

(Application for copies of ASTM Methods of testing may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractors (see 6.3). Variation from this specification may appear in the sample in which case this specification shall govern.

* 3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.2).

* 3.3 Materials. (see 6.4).

3.3.1 Basic material. The basic material shall conform to Class 2 of MIL-C-43191. The color of the material shall match OG-107.

* 3.3.1.1 Binding. The material for binding the neckline shall be made from basic material specified in 3.3.1, bias-cut on a 45 (± 5) degree angle in strips of 1 1/4 (± 1/8) inches wide. The ends of the strips shall be joined by seam type SSa-1 and stitch type 301 with 10 to 12 stitches per inch. The seams shall finish 1/8 to 3/16 inch wide and shall be spread open.

3.3.2 Vinyl laminated fabric. The vinyl laminated fabric shall conform to the following requirements when tested as specified in 4.4.1.

3.3.2.1 Cotton, jersey knit, fleece. The fabric shall be made from cotton, jersey knit and napped on one side to form an even fleece surface. It shall weigh 8.5 to 9.5 ounces per square yard. The color shall be white.

3.3.2.2 Film, flexible vinyl. The flexible vinyl film shall be formulated for resistance to cracking and blocking at -40°F, and have oil resistance. The film shall be a minimum of 15 mils in thickness and shall be pigmented to approximate the shade of the basic material.

3.3.2.3 Lamination. The flexible vinyl film shall be laminated to the smooth side of the cotton knitted fleece. The use of any adhesive shall be prohibited.

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* 3.3.2.4 Physical requirements. The laminated fabric shall conform to the physical requirements shown in Table I.

Table I - Physical requirements of laminated fabric

Characteristics	Requirements (min.)
Weight per square yard	23.5 ounces (797 g/m^2)
Thickness at 0.1 psi	0.090 inch (.23cm)
Thickness at 1.1 psi	0.070 inch (.18cm)
Hydrostatic pressure coating up	80 psi (548 pascal)
Hydrostatic pressure coating down	85 psi (582 pascal)
Adhesion (2 inch strip)	5 pounds (22 newtons)

* 3.3.3 Neoprene foam. The neoprene foam shall be an expanded chlorinated cellular rubber having a closed cell structure conforming to ASTM designation types SC-41. It shall be pigmented black, have natural skin on one or two sides, and shall conform to the physical requirements shown in Table II when tested as specified in 4.4.1.

Table II - Physical requirements of neoprene foam

Characteristics	Requirements	Tolerance
Thickness	1/4 inch	± 1/16 inch
Compression set	30 percent	Maximum
Density	10 (lbs. per $ft.^3$)	Minimum

* 3.3.4 Fastener, nylon, tape. The nylon tape fastener shall conform to Type II, Class 1 of MIL-F-21840. The hook portion for the left front closure and hook and pile for the leg closure shall be 1 inch wide. The pile portion for the right front closure shall be 2 inches wide. The color shall approximate the shade of the basic material (see 3.3.1).

* 3.3.5 Thread, polyester/cotton-covered. The polyester cotton covered thread shall conform to ticket numbers 30 and 50, 2 or 3 ply of MIL-T-43548. The color shall approximate the shade of the basic material (see 3.3.1).

* 3.3.6 Tape, cotton. The cotton tape for the front opening tabs and reinforcing armholes, gussets, waistline, crotch, outseams, inseams, and seat seam shall conform to Type I, classes 1, 2, or 3, 1 1/4 inches wide of DDD-T-86. If dyed tape is used, the color shall be O. G. 107.

3.3.7 Braid, cotton, tubular. The cotton braid for the waistline drawstrings shall conform to Type V, class 1 of MIL-B-371, except that colorfastness to laundering only shall apply. The color shall be O.G. 107.

3.3.8 Webbing, elastic. The elastic webbing for the sleeve hem channel shall be natural and shall conform to class 1, 3/4 inch wide of MIL-W-5664.

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3.3.9 Hardware.

* 3.3.9.1 Eyelets and washers. The eylets and washers for the braid in the waistline channel shall conform to eyelet number BBE-118 and washer number BBW-101 of MIL-E-20652/1 except that the overall height shall be 0.176 inch and the finish shall be as specified in 3.3.9.3.

3.3.9.2 Fasteners, snap. The snap fasteners for the front tabs and pocket flaps shall conform to style 2, 24-line, size 1 button with socket, and stud with eyelet size 1, 2 or 3 of MS 27980, except that the finish shall be as specified in 3.3.9.3.

3.3.9.3 Finish. The eyelets and snap fasteners shall have either a black chemical finish conforming to MIL-F-495, except that the requirements to gloss shall not apply, or a nickel plated finish conforming to class 2 of QQ-N-290, except that the minimum thickness shall be 0.0005 inch. The button cap of the snap fasteners shall have a baked-on enamel finish conforming to type I, or II of MIL-E-480. The color of the button cap shall approximate the shade of the basic material.

* 3.3.10 Labels

3.3.10.1 Identification and care label. Each coverall shall have a combination identification and care label conforming to Type I, Classes 1 and 3 of DDD-L-20. The printing shall show fastness to laundering. The label shall bear the following inscription:

COVERALLS, CATAPULT, CREWMAN'S
50% COTTON, 50% NYLON
CONTRACT NO. DLA-000-00-0-0000 (EXAMPLE)
NAME OF CONTRACTOR:

Hand or Machine Wash. Use warm water and mild soap or detergent. DO NOT USE CHLORINE BLEACH. Drip or tumble dry at moderate temperature not to exceed 160°F (70°C). Remove immediately.

3.3.10.2 Size label. Each coverall shall have a size label conforming to Type I, Class 2 of DDD-L-20. The printing shall show fastness to laundering. The label shall bear the following inscription:

SIZE: MEDIUM (40-42) (EXAMPLE)
STOCK NUMBER: 8415-00-000-0000 (EXAMPLE)

3.4 Description. The garment shall be a coverall style having a shirred yoke back and a front opening using a "hook" and "pile" nylon tape fastener closure with 3 snap fastener tabs on inside. An opening with nylon tape fastener closure is provided at side of each leg bottom. The coveralls shall have an inside drawcord waist adjustment and a patch pocket with snap fastener closure flap located on the right thigh. Reinforcement patches and pads made of vinyl laminated material and foam shall be located at each elbow and knee. Each sleeve bottom shall have an elastic take-up adjustment and gussets shall be located at each armhole and at the crotch.

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3.4.1 Figure. The figure is furnished for information only. Any inconsistancies between the specification and the figure, the specification shall govern.

3.5 Patterns. Standard patterns to be used for cutting the contractors working patterns will be furnished by the Government (see 6.3). The working patterns shall be identical to the standard patterns and shall not be altered in any way. The patterns allow 3/8 (+ 1/16) inch seam allowance for all joining seams and as otherwise indicated in Table III.

3.5.1 Pattern parts. The component parts of the coverall shall be cut from the material as specified below:

Material	Nomenclature of pattern parts	Cut parts
Cloth, cotton/nylon (Shirt)	Back	1
	Back yoke	2
	Front	2
	Sleeve	2
	Armhole gusset	2
	Elbow pad cover (inside protection)	2
(Trousers)	Back	2
	Front	2
	Crotch gusset	1
	Patch pocket (right side)	1
	Patch pocket reinforcement	1
	Patch pocket flap	1
	Knee pad cover (inside protection)	2
	Inside waist channel (waist adjustment)	1
Template	Patch, pressing pocket	-
Neoprene	Knee pad	2
	Elbow pad	2
Laminated vinyl fabric	Knee pad (outside protection)	2
	Elbow pad (outside protection)	2

3.6 Construction.

3.6.1 Stiches, seams, and stitching. Stitch, seam and stitching types specified in Table III shall conform to FED-STD-751. Wherever two or more methods for seam or stitch types are given for the same part of the operation, any one of them may be used. Where stitch type 401 is used, the looper or underthread shall be on the inside of the coveralls. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats, or open seams will result. Unless otherwise specified, top stitching and edge stitching shall be 1/16 inch from the folded edge.

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3.6.2 Thread breaks and ends of seams. Ends of all seams and stitchings when not caught in other seams or stitchings shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks (401 stitch type) shall be repaired by using 301 stitch type. Where stitch type 401 is permitted, as an option, it shall not be used unless the ends are caught in other seams or stitching.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table III.

3.7 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with operation requirements specified in Table III. The contractor is not required to follow the exact sequence of operations as specified, provided the finished coverall is identical to that produced by following the sequence of operations as specified.

3.7.1 Size marking. The component parts of the coveralls shall be marked, ticketed or bundled to insure a uniform size throughout the garment. Any method may be used except:

(1) Corrosive metal fastening devices.
(2) Sew-on shade tickets.
(3) Adhesive type tickets which discolor or adhere to the material upon removal of tickets.

NOTE: The use of ink pad numbering machines, rubber stamps or pencil is allowed, provided the numbering does not show on the outside and wherever possible, is covered by the seam allowance.

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Table III - Construction of coveralls

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting. a. The coveralls shall be cut in strict accordance with patterns furnished, which show size, shape and directional lines for cutting, and notches for proper assembly of all parts. The use of holding devices such as staples, etc., on the paper pattern parts which puncture the base fabric is prohibited.					
*	b. Drill marks for positioning any assembly are prohibited except for placement of markings for the eyelets on the waist channel and the snap fastener on the patch pockets.					
	c. Directional lines on the fronts and backs may vary from the warp direction by not more than 1 1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the directional lines to the selvage edge (on the even side of lay) of the fabric and the difference between the two measurements shall not exceed 1 1/2 inches.					
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in MIL-STD-1688 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Prepare and attach elbow and knee (inside and outside protection) pad covers.					
*	a. Overedge the four sides of the (inside protection) elbow and knee pad covers of basic fabric.	503 or 504 or 602	EFd-1	6-10	50	50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
3.	Prepare and attach elbow and knee (inside and outside protection) pad covers. (cont'd)					
	b. Stitch, 1/8 inch from overedged edge, face side up, the three sides of the (inside protection) elbow and knee pad covers on the underside as indicated by marks on pattern with the wider portion upward. Insert neoprene pad and stitch, 1/8 inch from overedged edge closing the fourth side. Cross stitch through all plies.	301	LSbj-1	10-12	30	30
	c. Double stitch, 1/4 inch gauge, the vinyl pads (outside protection) with the outer row 1/16 to 1/8 inch in from the edge. Care shall be taken that no neoprene edge is caught in the stitching.	301	LSa-2	10-12	30	30
*	d. Stitch combination identification/care label to the right front, on the inside of the trouser portion, under patch pocket. Label stitching shall not show on the outside of garment after the pocket is attached.	301	LSbj-1	10-12	30	30
4.	Prepare and attach pocket (right side).					
*	a. Fold and form a 1 (± 1/8) inch wide pocket hem with a 3/8 to 1/2 inch turn under. Stitch, 1/16 inch from folded edges, the top and bottom edges (Basting folder may be used to form pocket).	301 or 401	EFt-2	10-12	30 30	30 50
	b. Position, face side up, the pocket reinforcement on the outside with the top edge turned under 3/8 inch and the bottom and side edges even. Stitch, 1/16 inch from edge, the top folded edge, down side and across bottom edges.	301 or 401	LSd-1 and SSa-1	10-12	30 30	30 50

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Table III - **Construction** of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Prepare and attach pocket (right side) cont'd c. Attach snap fastener (stud and post) on pocket as indicated by marks on pattern with a self fabric reinforcement stay on the underside.					
*	d. Turn under 1/2 inch the side and bottom edges (with attached reinforcement piece) of the pocket. Pre-cresing or use of basting folder is permitted. Position pocket on right front of trousers as indicated by marks on patterns and double stitch 1/4 inch gauge with the outer row 1/16 inch from the turned edge.	301 or 401	LSd-2	10-12	30 30	30 50
	e. Fold flap in half lengthwise and stitch, 3/8 inch gauge, the side edges. Turn to finished position and stitch, 1/4 inch from folded edge, the side and folded edges.	301	SSe-2 and OSf-1	10-12	30	30
	f. Attach snap fastener (cap and socket) on flap to correspond with stud and post on pocket.					
	g. Set raw edge of flap 3/8 to 5/8 inch above pocket opening and stitch 1/4 inch from raw edge. Trim seam allowance, turn to finished position and stitch 1/4 inch from turned edge.	301	LSs-2	10-12	30	30
5.	Join yoke to shirt fronts and back. a. Shirr shirt back at top and waist edge between the notches.	301	OSa-1	10-12	30	30
	b. Join, 1/2 inch from raw edge, the shirt back between the straight edge of the two yokes (face side in).	301 or 401	SSa-1	10-12	30 30	30 50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
5.	Join yoke to shirt fronts and back (cont'd)					
	c. Join, 1/2 inch from raw edge, the shirt fronts between the yokes (face side in) at the shoulder. Turn yoke, back and fronts to finished position.	301 or 401	SSa-1	10-12	30 30	30 50
6.	Join trouser - shirt front waistline.					
*	a. Join, face to face, the trouser-shirt front waistline with the tape on the uppermost side extending 1/4 to 1/2 inch beyond the front edge and even at the side seam. Serge waistline.	301 or 401 and 503 or 504 or 602	SSab-1 SSab-1	10-12 6-10	30 30 50	30 50 50
	OR					
	b. As an alternate, waistline may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50
	c. Turn tape over joining seam with the free edge lapped onto the trouser side and edgestitch.	301 or 401	LSa-1	10-12	30 30	30 50
*	d. Join, face to face, the left and right trouser fronts at the crotch, below opening with the tape on the underside. The top edge of the tape shall be turned under with the stitching extending 1/2 inch beyond the folded edge. Serge front crotch.	301 or 401 and 503 or 504 or 602	SSab-1 SSab-1	10-12 6-10	30 30 50	30 50 50
	OR					
	e. As an alternate, crotch may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
6.	Join trouser - shirt front waistline (cont'd)					
	f. Turn tape over joining seam with the free edge lapped onto the left trouser front. Stitch, 1/16 inch from folded edge, across fold and down side edge of tape.	301 or 401	LSa-1 and LSd-1	10-12	30 30	30 50
7.	Prepare snap fastener closure tabs for left and right front opening.					
	a. Fold cotton tape for the right front opening in half across the length. Edgestitch the folded top and side edges. Center, 1/2 (± 1/8) inch from folded edge, and secure snap fastener (stud and post) through double ply of tab.	301 or 401	SSa-1 and OSf-1	10-12	30 30	30 50
	b. Fold cotton tape for the left front opening in half across the length. Center, 3/4 (± 1/8) inch from folded edge and secure snap fastener (cap and socket) through double ply of tab.					
	c. Length of closure tabs shall finish as follows: Right front 1 1/2 (± 1/4) inches Left front 1 1/4 (± 1/4) inches					
8.	Fold and form closure for right and left front opening.					
	a. Fold and form on underside of left front opening a 1 1/4 inch wide facing as indicated by notches on pattern. Position the 1 inch wide "hook" portion of the nylon fastener tape midway between edges of the facing and edgestitch, simultaneously, inserting and catching at the inner edge, three closure tabs with the stud and post portion of the snap fastener facing	301	EFq-2 (except strip outside)	10-12	30	30

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Fold and form closure for right and left front opening (cont'd) upward. Back edge of facing shall be turned under and caught between the plies and double row of stitching.					
	b. To facilitate operation "a" inner edge of facing may be pre-formed.	301 or 401	EFa-1	10-12	30 30	30 50
	c. Edgestitch the free edges of closure taps. Stitching shall not extend beyond the nylon fastener tape.	301	LSa-1	10-12	30	30
	d. Connect, 1/16 inch from edge, the double row of stitching joining the nylon fastener tape at the bottom edge.	301	LSa-1	10-12	30	30
	e. Fold right front opening 1/2 to 5/8 inch on the face side. Position the 2 inch wide "pile" portion of the nylon fastener tape 1/16 inch in from the folded edge and edgestitch the outer edge simultaneously inserting and catching the waistline seam tape and three closure tabs between the plies, with the cap portion of the snap fastener facing downward. Edgestitch the inner and bottom edges.	301 or 401	SSt-2	10-12	30 30	30 50
	f. Stitch bottom of left and right front opening, through all plies, the width of the left front facing connecting the two rows of stitching.	301	LSa-1	10-12	30	30
	g. Tabs on the left front shall be located as follows: - One, 5/8 (± 1/8) inch from edge of neckline. - One, 1 (± 1/8) inch from waistline joining seam. - One, equidistant between the two.					

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
8.	Fold and form closure for right and left front opening (cont'd)					
	h. Tabs on right front shall correspond with tabs on left front.					
9.	Join trouser - shirt back.					
*	a. Join, face to face, the seat seam with the tape on the uppermost side. Serge seat seam.	301 or 401 and	SSab-1	10-12	30 30	30 50
		503 or 504 or 602	SSab-1	6-10	50	50
	OR					
	b. As an alternate, seat seam may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50
	c. Turn tape over joining seam with the free edge lapped onto left trouser back and edgestitch.	301 or 401	LSa-1	10-12	30 30	30 50
*	d. Join, the trouser - shirt back waistline with the tape on the uppermost side. Serge waistline.	301 or 401 and	SSab-1	10-12	30 30	30 50
		503 or 504 or 602	SSab-1	6-10	50	50
	OR					
	e. As an alternate, waistline may be joined as one operation with a safety stitch.	516 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
9.	Join trouser - shirt back (cont'd)					
	f. Turn tape over joining seam with the free edge lapped onto trouser back and edgestitch.	301 or 401	LSa-1	10-12	30 30	30 30
10.	Bind neckline					
	a. The binding shall measure 3/8 (± 1/16) inch wide with raw edges securely caught in the binding. Not more than one splice shall be permitted.					
	b. Stitch, 3/8 inch from raw edge, the binding to the neckline. Turn and fold edge and ends and edgestitch.	301	BSj-2	10-12	30	30
	OR					
	c. As an alternate, neckline may be bound with a folder attachment.	301	BSc-1	10-12	30	30
	d. To facilitate operation "b" and "c" yoke neckline may be pre-stitched.	301 or 401	SSa-1	10-12	30 30	30 50
11.	Make and attach hanger loop.					
	a. The hanger loop shall be made from basic material and formed with a covering stitch on the underside or by folding in half lengthwise, turning in raw edges parallel to the fold and edgestitch.	406 or 301 or 401	EFh-1 EFp-2	10-12	30 30 30	30 30 50
	b. Center the hanger loop on the inside of yoke at base of binding with ends equidistant between the shoulder seams. Turn ends under and tack or bartack through all plies, across the width 1/8 to 3/16 inch in from the folded edge.	301 or Bartack	LSd-1 LSd-1	10-12 24-42 per bartack	30 50	30 50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
11.	Make and attach hanger loop. (cont'd)					
	c. The finished hanger loop shall measure 3/8 (± 1/8) inch wide and 2 1/2 (± 1/4) inches between the stitching.					
*12.	Attach size label Edgestitch the four sides of the size label onto the yoke directly below the hanger loop.	301	LSbj-1	8-10	30	30
13. *	Join trouser inseams. a. Join, face to face, the inseams with the tape on the underside. Serge inseams.	301 or 401 and 503 or 504 or 602	SSab-1	10-12	30	30
					30	50
	OR		SSab-1	6-10	50	50
	b. As an alternate, the inseams may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50
	c. Turn tape over joining seam with free edge lapped onto the left and right trouser back and edgestitch.	301 or 401	LSa-1	10-12	30 30	30 50
14. *	Join shirt - trouser outseams. a. Join, face to face, the outseams with the tape folded under at the leg opening and on the underside. Serge outseams.	301 or 401 and 503 or 504 or 602	SSab-1	10-12	30	30
					30	50
	OR		SSab-1	6-10	50	50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
14.	Join shirt - trouser outseams (cont'd)					
	b. As an alternate, outseams may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) (overedge)	50
	c. Turn tape over joining seam with the free edge lapped onto the shirt - trouser back and edgestitch	301 or 401	LSa-1and LSd-1	10-12	30 30	30 50
15.	Fold and form closure for leg opening.					
	a. Fold and form on underside of leg opening a 1 (± 1/4) inch wide hem with a 1/4 inch turn under. Stitch 1/16 inch from folded edge.	301 or 401	EFb-1	10-12	30 30	30 50
	b. Fold and form on underside of front leg opening a 1 1/4 (± 1/8) inch wide facing as indicated by notches on pattern. Position the 1 inch wide "hook" portion of the nylon fastener tape midway between edges of facing. Edgestitch the four sides.	301 or 401	EFq-2 (except outside strip)	10-12	30 30	30 50
	c. Fold back leg opening 3/8 inch to 1/2 inch on face side. Position the 1 inch wide "pile" portion of the nylon fastener tape 1/16 inch in from the folded edge and edgestitch the four sides.	301 or 401	SSt-2	10-12	30 30	30 50
	d. Stitch top edge of leg opening, through all plies, the width of the formed facing.	301 or 401	LSa-1	10-12	30 30	30 50
16. *	Close and attach sleeves.					
	a. Join, face to face, the sleeve inseam with the tape on the underside. Serge inseam.	301 or 401 and 503 or 504 or	SSab-1 SSab-1	10-12 6-10	30 30 50	30 50 50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
16.	Close and attach sleeves (cont'd)					
	OR					
	b. As an alternate, sleeve inseams may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50
	c. Turn tape over joining seam with the free edge lapped onto the back portion of the sleeve and edgestitch.	301 or 401	LSa-1	10-12	30 30	30 50
*	d. Stitch, 3/8 inch from raw edge, face to face, the sleeve to the armhole with the tape on the uppermost side. Serge armhole.	301 or 401 and 503 or 504 or 602	SSab-1	10-12	30 30 50	30 50 50
	OR					
	e. As an alternate, armhole may be joined as one operation with a safety stitch.	515 or 516 or 519	SSab-2	10-12	30 (chain) 50 (overedge)	50 50
	f. Turn tape over joining seam with the free edge lapped onto the front and back body and edgestitch.	301 or 401	LSa-1	10-12	30 30	30 50
	g. Overlap 1/2 inch and stitch 1/4 inch from raw edge, the ends of the elastic webbing.	301	LSa-1	10-12	30	30

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Table III - Construction of coveralls

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
16.	Close and attach sleeves. (cont'd)					
	h. Fold and form a hem 1 1/4 (± 1/8) inches wide as indicated by notches on pattern. The raw edge shall be turned under 1/4 inch and the elastic webbing inserted and caught within the fold. Care shall be taken that the webbing is not caught in the stitching.	301	EFr-1	10-12	30	30
17.	Attach armhole and crotch gussets.					
*	a. Stitch, 3/8 inch from raw edge, face to face, the armhole gusset to the body. Serge armhole gusset.	301 or 401 and	SSa-1	10-12	30 30	30 50
	OR	503 or 504 or 602	SSa-1	6-10	50	50
	b. As an alternate, armhole gusset may be joined as one operation with a safety stitch.	515 or 516 or 519	SSa-2	10-12	30 (chain) 50 (overedge)	50 50
	c. Bind armhole gussets with the overlapping end turned under, caught and concealed on the underside. Position the binding flat and stitch, 1/16 inch from edge the outer edge to the body.	301 or 401	BSc-1 and LSa-1	10-12	30 30	30 50
*	d. Stitch, 3/8 inch from raw edge, face to face, the crotch gusset to the body. Serge crotch gusset.	301 or 401 and	SSa-1	10-12	30 30	30 50
	OR	503 or 504 or 602	SSa-1	6-10	50	50
	e. As an alternate, crotch gusset may be joined as one operation with a safety stitch.	515 or 516 or 519	SSa-2	10-12	30 (chain) 50 (overedge)	50 50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
17.	Attach armhole and crotch gussets (cont'd) f. Bind crotch gusset with overlapping end turned under, caught and concealed on the underside. Position the binding flat and stitch, 1/16 inch from edge, the outer edge to the body.	301 or 401	BSc-1 and LSa-1	10-12	30 30	30 50
18.	Attach waist channel strip. a. Waist channel strip shall be centered over the inside waistline seam with the finished ends extending to the inner edges of the nylon fastener tape. There shall be two drawstrings (cotton braid) of the following lengths. Size — Length (inches) — Tolerance (inches) Small — 42 — ± 1/2 Medium — 46 — ± 1/2 Large — 50 — ± 1/2 b. Center the drawstrings (cotton braid) on the underside 1 (± 1/8) inch in from each end of the waist channel strip. Secure each end of the drawstrings with two 3/8 to 1/2 inch vertical bartacks, 1/4 inch apart. c. Attach eyelets as indicated by marks on pattern with a self fabric reinforcement stay on the underside. d. Insert drawstrings (cotton braid) through eyelets and knot ends.	Bartack		28-42 per bartack	50	50

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Table III - Construction of coveralls (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
18.	Attach waist channel strip (Cont'd)					
	e. Stitch, 1/16 to 1/8 inch from folded edge, the four sides of the waist channel strip with a 1/4 to 3/8 inch turn under.	301	LSd-1	10-12	30	30
19.	Bartacking. All bartacks shll be horizontal 3/8 to 1/2 inch wide and located as follows: - One, at each end of the pocket opening, 1/8 to 3/16 inch below top edge of hem fold connecting the double row of stitching. Bartack shall not extend beyond the side edges. - One, centered on outseam, at top edge of leg opening superimposed on line of stitching. - One, centered on front crotch seam, at bottom edge of front opening superimposed on line of stitching.	Bartack		28-42 per bartack	50	50
20.	Cleaning. Trim all ends of stitching and remove all loose threads, spots, and stains and all size markings.					
21.	Examine, close and pack coveralls. a. Examine coveralls for correct labeling, assembly, material imperfections, workmanship damages, cleanness and dimensions. b. Secure front, leg and pocket closures. c. Fold and pack coveralls as specified in Section 5.					

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3.8 Sizes and measurements. The sizes and measurements of the finished coveralls shall be as shown in Table IV. All measurements and tolerances are expressed in inches.

Table IV - Sizes and measurements

Size	Chest A	Waist B	Hip C	Length D	Leg Inseam E	Sleeve Inseam F
Small (36-38)	42	35	43	59 1/2	28 1/2	20 1/2
Medium (40-42)	46	39	47	63	30 1/2	22 1/2
Large (44-46)	50	43	51	66 1/2	32 1/2	24 1/2
Tolerance	± 1	± 1	± 1	± 1	± 3/4	± 1/2

NOTE: A, B, C, D, E and F refer to Figure 1.

A. Chest - Twice the measurement across the chest at the bottom of the armhole gusset.

B. Waist - Twice the measurement taken across the waist seam.

C. Hip - Twice the measurement taken across the base of the front fly opening.

D. Length - Taken from fold of neck at shoulder to bottom of trousers.

E. Leg Inseam - Taken from center of crotch gusset to bottom of the trousers.

F. Sleeve inseam - Taken 2 inches above bottom edge of gusset to bottom of sleeve.

3.9 Workmanship. The finished coverall shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable "point values" or defect limit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified by the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. The first article submitted in accordance with 3.2 shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship and dimensional requirements.

* 4.4 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-1668.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, components listed in Table V shall be tested for the characteristics specified and in accordance with the referenced test method of FED-STD-191, unless otherwise specified. All test reports shall contain the individual values utilized in expressing the final result. Unless otherwise specified in subsidiary specifications, sample size shall be in accordance with the following:

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The unit for expressing lot sizes and sample unit for testing each component shall be in accordance with applicable subsidiary specifications and as follows:

Component	Lot size expressed as	Sample unit for testing
Cloth, cotton, jersey knit, fleece	Yards	3 yards full width
Film, flexible vinyl	Sheet	6 sheets
Laminated fabric	Yards	3 yards full width
Neoprene foam	Sheet	6 sheets
Coating (baked-on enamel) fastener, snap (button cap)	Gross	6 of each enamel finish

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Table V - Testing of components

Component	Characteristic	Requirement paragraph	Test method
Cloth, cotton, jersey knit, fleece	Material identification	3.3.2.1	1200 1/
	Knit	3.3.2.1	Visual
	Napped fleece	3.3.2.1	Visual
	Weight	3.3.2.1	5041
	Color	3.3.2.1	Visual
Film, flexible vinyl	Material identification	3.3.2.2	1/
	Resistance to cracking and flaking at -40¼F	3.3.2.2	4.4.1.1 1/
	Oil resistance	3.3.2.2	4.4.1.2 1/
	Thickness	3.3.2.2	1/
	Color	3.3.2.2	Visual
Laminated fabric	Weight	3.3.2.4	5040
	Thickness	3.3.2.4	5030 2/
	Hydrostatic	3.3.2.4	5512
	Adhesion	3.3.2.4	5950 3/
Neoprene foam	Material identification	3.3.3	D-1056 (ASTM)
	Color	3.3.3	Visual
	Thickness	3.3.3	FED-STD-601, 12031
	Compression	3.3.3	FED-STD-601, 12131 4/
	Density	3.3.3	See 4.3.1.3
Fasteners, snap (button cap)	Enamel finish	3.3.9.3	MIL-E-480 1/

1/ A certificate of compliance will be acceptable for this requirement.

2/ Except that the pressure shall be at 0.1 psi and 1.1 psi.

3/ Tested in the wale direction only and the distance between the clamps shall be 1/2 inch at the start of the test.

4/ Except that the specimen shall be 1/2 inch thick and shall be tested at room temperature.

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4.4.1.1 Cracking resistance. A sample of vinyl film not less than 8 by 8 inches shall be exposed in a cold chamber of -40¼F (± 3.6¼F) for 4 hours. Without being removed from the cold chamber, the sample shall be creased 180¼ in the center by folding slightly while lying on a flat, smooth surface. A steel roller, approximately 5 1/2 inches in diameter and 2 inches wide and having a weight of approximately 10 pounds shall be rolled across the crease line one time. The roller shall be at the same temperature as the sample. The sample shall be handled with gloves and care shall be taken so that its temperature remains uniform throughout the test. The sample shall then be folded open and visually inspected for signs of cracking or flaking. Any evidence of cracking or flaking constitutes a failure.

4.4.1.2 Oil resistance. An 8 by 8 inch sample of vinyl film shall be placed on a wood frame. The inside dimensions of the wood frame shall be 6 by 6 by 1 inch. The sample shall be forced into the frame by a wood block 5 3/4 by 5 3/4 inches (with rounded corners) to form a basin of uniform depth. The edges of the vinyl film shall be tacked to the frame and the block removed. Lubricating oil, conforming to MIL-H-5606, shall be rapidly poured into the basin to a 1/2 inch depth. After the oil has been in the basin for one hour, the bottom of the vinyl film sample forming the basin shall be examined to determine if oil has permeated the film to cause leakage. Any evidence of leakage constitutes a failure.

4.4.1.3 Density. Five (5) specimens each 2 inches square shall be weighed separately to the nearest 0.001 gram on an analytical balance. The thickness of each specimen shall be determined in accordance with Test Method 5030 of FED-STD-191, except that the pressure shall be at 0.01 psi. The following formula shall be used to determine density.

$$D = \frac{0.95 \times W}{T}$$

Where: D = Density in pounds per cubic foot
W = Weight of sample in grams
T - Thickness of sample in inches

* 4.4.2 End item examination. Sampling and examination of the end item shall be as specified in MIL-STD-1688.

* 4.4.3 Packaging inspection. Packaging inspection shall be in accordance with MIL-STD-1668.

5. PACKAGING

5.1 Preservation-packaging. Preservation packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A. Each coverall shall be folded as follows:

1. Secure closures and snap fasteners.
2. Lay garment flat, face down.
3. Fold garment in half, lengthwise, with the crotch inside the fold.
4. Fold sleeves over body with armhole gussets inside the fold and the sleeve at the center of the folded garment.
5. Bring bottom of trousers over knee patch.
6. Using the top of the knee patch as a guide, fold twice across the length of the garment.
7. Fold top of garment over the previous folds.

The folded garment shall measure approximately 11 1/2 inches by 15 1/2 inches. Each folded coverall shall be inserted in a snug-fitting flat style clear polyethylene film bag of 0.0015 inch thickness (± 20% tolerance). All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. As an alternate, the polyethylene bag may be of the tuck-in or reverse flap type, in which a heat seal closure and corner vent hole are not required.

5.1.2 Level C (Commercial packaging). Coveralls shall be packed to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four (4) coveralls of one size only, packaged as specified in 5.1, shall be packed within a fiberboard shipping container assembled, closed and reinforced conforming to class weather-resistant, grade V2s, type CF, variety SW size 2A of MIL-B-17757. The fiberboard for the liner shall conform to Type CF, class weather-resistant, variety DW, grade V15c of MIL-B-17757. Level A packages shall be packed flat within the shipping container as follows: Two tiers of two packages with each coverall reversed and the length of the coveralls parallel to the side panels of the container. Toward the end of the contract or order, or when there is less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

5.2.2 Level B. Four (4) coveralls of one size only, packaged as specified in 5.1, shall be packed within a fiberboard shipping container, assembled and closed, conforming to class domestic, grade 275, type CF, variety SW size size 2A of MIL-B-17757. The fiberboard for the liner shall conform to Type CF, class domestic, variety DW, grade 275 of MIL-B-17757. Level A packages shall be packed flat within the shipping container as follows: Two tiers of two packages with each coverall reversed and the length of the coveralls parallel to the side panels of the container. Toward the end of the contract or order, or when there is less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

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5.2.2.1 When specified, (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

5.2.3 Level C (Commercial packing). Coveralls, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Polyethylene bagged packages. Polyethylene bagged packages specified in 5.1.1, shall have the required information legibly printed or stamped in black bold letters, 1/4 inch in height, directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
SIZE
QUANTITY

5.3.2 Labels, mixed sizes. Each shipping container, packed with mixed sizes, shall have securely attached to the end and side, directly under the printing and stenciling, a white paper label, 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

6. NOTES

6.1 Intended use. The coveralls covered by this specification are intended for wear by carrier flight deck personnel engaged in launching aircraft by means of a catapult.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Size required (see 1.2).
c. First article requirements (see 3.2).
d. Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

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6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids (see 3.1 and 3.5).

6.4 Recycled material. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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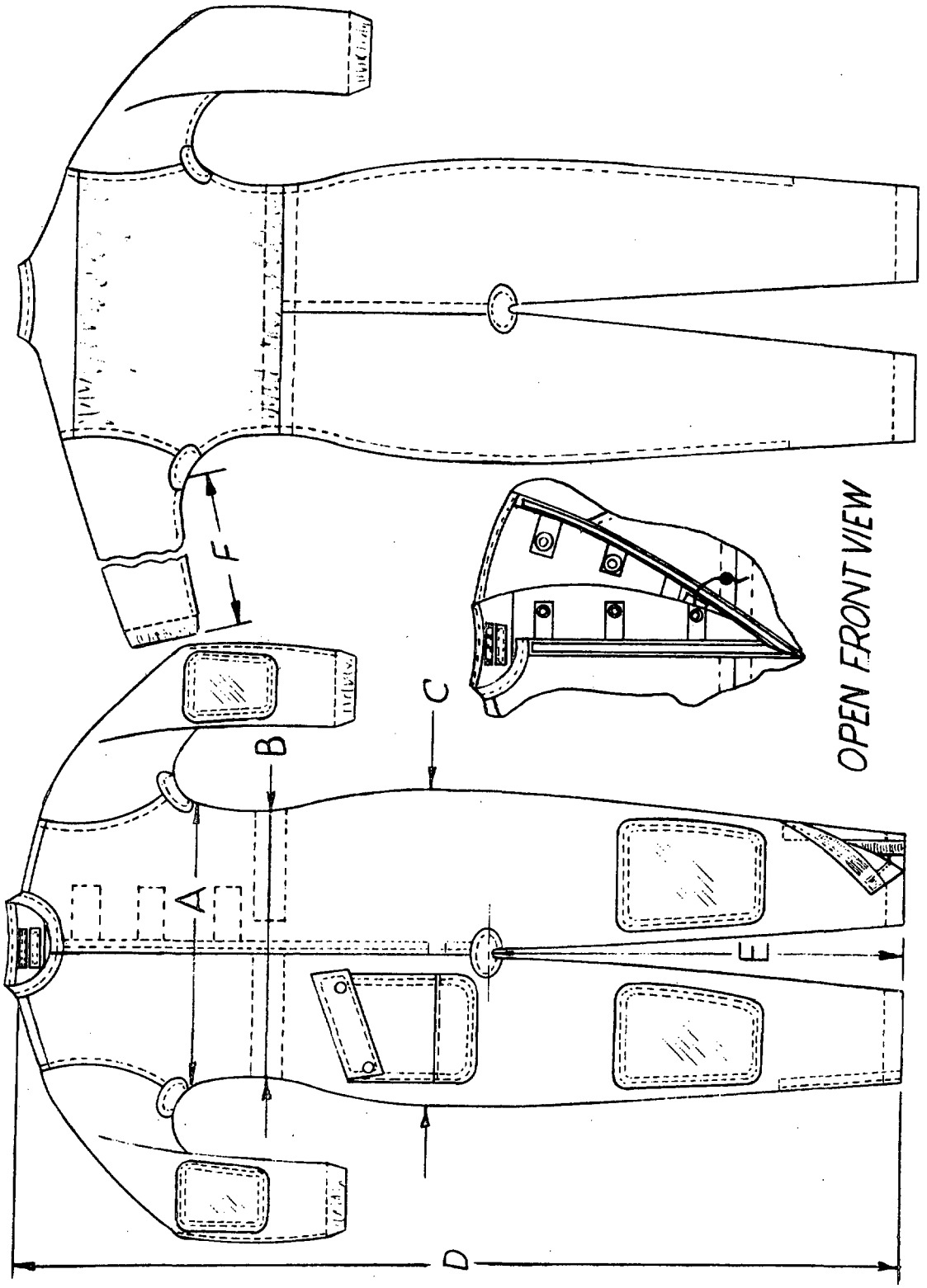


FIGURE 1 - COVERALLS, CATAPULT, CREWMEN'S

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